
Cross-Model Interpretability via Sparse Feature Alignment between BERT and GPT-2

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Abstract

Transformer-based classifiers achieve strong predictive performance but remain difficult to interpret due to distributed internal representations. This work investigates whether sparse autoencoder (SAE) features extracted from pretrained language model activations can serve as an interpretable intermediate representation that relates to downstream classification behavior. We fine-tune a BERT-based classifier on the IMDB sentiment classification dataset to obtain document-level predictions. To analyze internal structure, we extract residual stream activations from a pretrained GPT-2 model and decompose them into sparse latent features using pretrained sparse autoencoders (SAE Lens). These features are evaluated using three complementary statistical criteria: variance-based class separation, Welch’s t-test, and logistic regression coefficients. A composite ranking is used to identify task-relevant latent dimensions. Our results show that while a large fraction of SAE features exhibit statistically significant class separation, agreement between different importance metrics is limited, indicating a distributed notion of feature relevance. Qualitative analysis further suggests that strongly aligned features correspond to higher-level evaluative and semantic discourse, while weakly aligned features capture more stylistic, affective, or templated patterns. These findings suggest that SAE representations encode partially structured semantic information that is related to, but not reducible to, classifier decision boundaries.

1 Introduction

Large language models and transformer-based architectures have achieved remarkable success across a wide range of natural language processing tasks, including text classification, question answering, and generative reasoning (Vaswani et al. [2017], Devlin et al. [2019], Brown et al. [2020]). Despite their strong empirical performance, the internal mechanisms by which these models represent and process information remain poorly understood. Their hidden representations are typically distributed across high-dimensional activation spaces (Belinkov [2022]), making it difficult to identify which internal components correspond to meaningful semantic concepts or contribute to downstream predictions.

This challenge has motivated growing interest in mechanistic interpretability, a research area focused on understanding neural networks through analysis of their internal computational structure. A central obstacle in this area is the phenomenon of feature superposition (Elhage et al. [2022]), in which multiple latent concepts are encoded across overlapping activation dimensions. Because of this distributed encoding, individual neurons or hidden dimensions often do not correspond cleanly to interpretable concepts.

Sparse autoencoders (SAEs) have recently emerged as a promising approach for addressing this problem (Cunningham et al. [2023], Bricken et al. [2023]). By projecting dense transformer activations into a higher-dimensional sparse latent space, SAEs can disentangle overlapping representations into more interpretable features. Prior work has shown that sparse latent dimensions often correspond to meaningful semantic, syntactic, or conceptual structures within transformer residual streams (Cunningham et al. [2023], Bricken et al. [2023], Gurnee and Tegmark [2023]).

However, an important open question remains: to what extent do these sparse latent features align with downstream task behavior? While SAE features may be interpretable in isolation, it is less clear whether they capture information relevant to decision boundaries learned by supervised models (Marks and Tegmark [2023]).

This work addresses this question by introducing a cross-model analysis framework. We fine-tune a BERT-based classifier on the IMDB sentiment classification dataset (Maas et al. [2011]) and obtain document-level predictions on held-out examples. In parallel, we extract residual stream activations from a pretrained GPT-2 model using same held-out examples and decompose them using pretrained sparse autoencoders.

We then analyze whether SAE feature activations correlate with classifier predictions using multiple statistical measures, including variance-based separation, t-test, and logistic regression. In addition, we perform qualitative inspection of highly activating documents to characterize the semantic nature of aligned and non-aligned features. So in this work, we ask the following research question (RQ):

(RQ) *Do sparse autoencoder features extracted from GPT-2 residual streams align with BERT classifier predictions, and if so, what type of semantic structure do aligned features represent?*

We hypothesize that strongly aligned SAE features capture higher-level semantic and evaluative discourse, whereas weakly aligned features capture more stylistic, affective, or low-level linguistic patterns. However, we expect this relationship to be graded rather than binary.

The contributions of this work are as follows:

- We propose a cross-model framework connecting SAE-based interpretability with supervised classification behavior.
- We show that SAE feature importance exhibits weak but statistically significant alignment with classifier predictions, but varies substantially across ranking criteria.
- We provide qualitative evidence that strongly aligned features correspond to structured semantic and evaluative discourse, while weakly aligned features capture more affective and stylistic patterns.

Overall, our results suggest that SAE features encode partially structured semantic information that overlaps with but does not directly determine classifier decision boundaries.

2 Related Work

2.1 Interpretability in Transformer Models

Despite their strong empirical performance across natural language processing tasks, transformer-based models remain difficult to interpret due to the highly distributed nature of their internal representations. Unlike traditional feature-engineered systems, transformer activations emerge through large-scale optimization and often encode multiple overlapping concepts within the same activation subspace. This distributed nature is often analyzed through the lens of the Linear Representation Hypothesis (Park et al. [2023]), which posits that semantic concepts are encoded as directions in high-dimensional space rather than single neurons.

This challenge has motivated growing interest in mechanistic interpretability, a line of work focused on reverse-engineering internal computational structures of neural networks. A foundational contribution in this area is Toy Models of Superposition by Elhage et al. [2022], which formalized the phenomenon of feature superposition and demonstrated how neural networks compress multiple concepts into shared representational dimensions. Their work provides theoretical grounding for why individual neurons often fail to correspond to human-interpretable concepts.

Subsequent work has extended mechanistic interpretability to large transformer models by identifying circuits responsible for specific behaviors. For example, Ge et al. [2024] introduced methods for scalable circuit discovery in transformer architectures. These studies collectively suggest that understanding transformer behavior requires moving beyond neuron-level inspection toward feature-level decomposition.

2.2 Sparse Autoencoders for Feature Discovery

Sparse autoencoders (SAEs) have recently emerged as one of the most promising approaches for disentangling internal transformer representations into interpretable latent features.

A major milestone in this direction is (Cunningham et al. [2023]), which demonstrated that sparse dictionary learning can recover semantically meaningful latent features from transformer residual streams, addressing the problem of polysemantic neurons. Building on this, Kissane et al. [2024] extended sparse feature decomposition beyond residual streams to attention outputs, showing that SAE-based methods generalize across internal transformer components. Recent large-scale evaluations by OpenAI (2024) have further validated that SAEs can scale to frontier-class models while maintaining feature sparsity and high downstream reconstruction fidelity, establishing them as a robust tool for model-agnostic feature extraction Gao et al. [2024]. Further advancing circuit-level analysis, Dunefsky et al. [2024] proposed sparse approximations of transformer MLP computations that enable more scalable feature-circuit tracing.

These works establish that sparse feature decomposition can reveal semantically coherent latent concepts inaccessible through direct neuron inspection.

2.3 Sparse Representations and Interpretability Evaluation

Although sparse autoencoders have shown strong promise, evaluating whether discovered features meaningfully relate to downstream model behavior remains an open problem.

Much prior work focuses primarily on intrinsic interpretability metrics such as reconstruction loss, sparsity, and qualitative coherence of activations. For example, Shu et al. [2025] reviews current SAE methodologies and highlights the need for stronger behavioral validation of discovered sparse features. Similarly, O’Neill and Bui [2024] emphasizes the importance of connecting sparse latent structure to concrete functional behaviors.

While intrinsic metrics are valuable, the disconnect between interpretability and utility remains a critical bottleneck. USAEs (Lan et al. [2024]) have begun to explore universal feature spaces, but their work primarily focuses on feature similarity across vision models rather than the functional alignment of linguistic features with supervised decision boundaries.

These studies point to a key limitation in current SAE research: while sparse features are often interpretable in isolation, their relationship to task-specific predictive behavior is less frequently quantified.

2.4 Cross-Model Interpretability

Most prior SAE-based interpretability work analyzes sparse features within the same model from which activations are extracted. Comparatively little work examines whether sparse latent representations from one model align with decision behavior learned by another architecture. This distinction is important because cross-model agreement offers stronger evidence that sparse features capture generalizable semantic structure rather than model-specific artifacts.

Our work addresses this gap by correlating sparse features extracted from pretrained GPT-2 residual streams with sentiment predictions generated by a fine-tuned BERT classifier. Rather than interpreting features solely through intrinsic activation patterns, we evaluate their behavioral relevance using statistical association with downstream classifier outputs.

Our work aligns with the Platonic Representation Hypothesis (Huh et al. [2024]), which suggests that different models, despite varying architectures and training objectives, often converge toward a shared representational geometry of the world. By projecting GPT-2 activations into an SAE space and correlating them with BERT’s sentiment boundaries, we move beyond *interpretability-as-*

description toward *interpretability-as-validation*, testing whether these converged representations are truly task-relevant

This cross-model perspective enables a more stringent test of whether sparse latent dimensions correspond to semantically meaningful concepts relevant for supervised language understanding tasks.

3 Methodology

3.1 Classification Model Fine-Tuning

To establish a supervised classification baseline, a pretrained BERT-based transformer model was fine-tuned on the target text classification dataset (See table 1). The objective of this stage was to learn document-level predictive representations that could later serve as anchors for sparse feature interpretation. Input documents were tokenized using the corresponding BERT tokenizer and truncated to a fixed maximum sequence length. The classifier was optimized using cross-entropy loss with the AdamW optimizer (See figure 1).

Parameter	Value
Model	bert-base-uncased
Learning rate	2×10^{-5}
Batch size	32
Epochs	10
Sequence Length	128

Table 1: Training Configuration of BERT

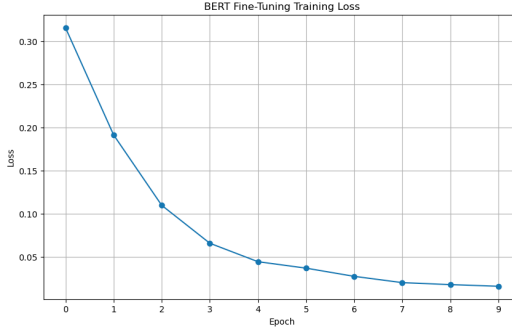


Figure 1: Training Loss During BERT Fine-Tuning

After training, the classifier generated predictions for the evaluation set. These predictions are used to define the target variable for downstream sparse feature analysis, allowing us to study how SAE features align with model-derived decision boundaries.

3.2 Sparse Autoencoder-Based Representation Decomposition

To interpret predictive behavior at the representational level, this work employs sparse autoencoders (SAEs) as a mechanism for decomposing dense transformer activations into interpretable latent features. A key challenge in mechanistic interpretability is that transformer hidden states are highly distributed: individual dimensions often participate in representing multiple overlapping concepts, a phenomenon commonly referred to as feature superposition.

Sparse autoencoders address this by projecting dense activations into a higher-dimensional latent space under sparsity constraints. Given a residual activation vector $h \in \mathbb{R}^d$, The encoder computes a sparse latent representation:

$$z = \text{ReLU}(W_e h + b_e)$$

where:

- W_e is the encoder projection matrix
- b_e is the encoder bias
- z is the sparse latent code

The decoder reconstructs the original activation:

$$\hat{h} = W_d z + b_d$$

Training enforces that only a small subset of latent dimensions activate for any given input. This sparse decomposition provides two major interpretability advantages: Latent dimensions often exhibit selective activation patterns that can correspond to semantically or functionally coherent features in the input distribution, and Individual features can be analyzed independently through activation-based inspection. This project uses pretrained sparse autoencoders provided through SAE Lens. The experimental configuration is, Base language model: GPT-2, SAE release: gpt2-small-res-jb, Residual extraction layer: 8, and Latent dimensionality: 24,576.

This configuration enables analysis of a high-capacity sparse feature dictionary while remaining computationally tractable. The objective is to determine whether these sparse latent dimensions capture interpretable concepts that correlate with classifier behavior. We focus on GPT-2 residual stream representations and BERT-based predictions to investigate whether different pretrained transformer models encode shared semantic structure that can be recovered through sparse feature decomposition.

3.3 Residual Stream Extraction

Each document is independently processed through GPT-2 to obtain intermediate residual stream activations. For a document of length T , the extracted hidden representation at layer l is

$$H^{(l)} \in \mathbb{R}^{T \times d}$$

where:

- T denotes token sequence length
- d denotes residual stream dimensionality

Residual activations are extracted specifically from layer 8. This layer was selected as an intermediate representational stage because this layer is commonly used in probing studies as it captures intermediate representations that reflect both lexical and higher-level contextual information. Unlike standard token-level analysis pipelines, document boundaries are explicitly preserved. This is necessary because downstream supervision is defined at the document level. Flattening activations across the corpus would destroy the correspondence between sparse features and classifier predictions.

3.4 Document-Level Sparse Representation Construction

Each token-level residual vector is encoded through the sparse autoencoder, producing token-level latent activations:

$$z_t \in \mathbb{R}^F$$

where $F = 24,576$.

To align these token-level activations with document-level labels, mean pooling is applied:

$$z_{\text{doc}} = \frac{1}{T} \sum_{t=1}^T z_t$$

This aggregation produces a single sparse feature vector for each document.

The resulting activation matrix is

$$Z \in \mathbb{R}^{N \times F}$$

where:

- N is the number of documents

- F is the number of sparse latent features

This matrix serves as the primary input for feature importance analysis.

3.5 Statistical Feature Ranking

The central analytical objective is to identify sparse latent features most strongly associated with classifier predictions. To ensure robustness, feature importance is evaluated using three complementary statistical criteria.

Variance-Based Class Separation For each feature, mean activation is computed separately for each predicted class. Importance is estimated as the variance across class-wise means. This measures the degree to which a feature discriminates between classes.

Welch’s t-Test Welch’s t-test evaluates whether activation differences between classes are statistically significant. Unlike the standard Student t-test, Welch’s formulation does not assume equal variance. This provides a more reliable estimate under heterogeneous activation distributions.

Logistic Regression Coefficient Analysis A logistic regression classifier is trained using sparse activations as predictors. The absolute magnitude of learned coefficients serves as a measure of predictive relevance. Features with larger coefficients contribute more strongly to class separation.

Composite Importance Scoring To combine complementary evidence, feature rankings are integrated into a unified score:

$$\text{Score} = \text{Variance} + |\beta| - \log(p)$$

where:

- Variance captures activation separability
- $|\beta|$ captures predictive contribution
- p captures statistical significance

This composite metric reduces reliance on any single criterion and improves robustness of feature selection. Top-ranked features (Strong SAE) and bottom-ranked features (Weak SAE) are then selected for qualitative interpretation through inspection of highly activating documents.

3.6 Qualitative Feature Interpretation

To interpret the semantic role of selected sparse features, a manual qualitative analysis was performed on documents exhibiting maximal activation for each feature.

For each top-ranked and bottom-ranked latent feature, the $k = 5$ highest-activating documents were retrieved from the evaluation corpus. These documents were manually examined to identify recurring linguistic, semantic, and structural patterns. Feature interpretation followed an inductive coding procedure:

- Inspect highest-activation documents
- Identify repeated semantic or syntactic patterns
- Assign provisional conceptual labels
- Compare labels across highly and weakly correlated features
- Refine labels through iterative consensus-building

Features were categorized into the following interpretive classes:

- Topic-semantic features (domain concepts, subject matter, thematic vocabulary)
- Sentiment/evaluative features (positive/negative polarity, affective framing)

- Stylistic features (tone, emphasis, rhetorical construction)
- Grammatical/syntactic features (function words, structural patterns)
- Mixed / ambiguous features

The analysis tests the hypothesis that features strongly correlated with classifier outputs encode semantically meaningful topical concepts, whereas weakly correlated features capture lower-level linguistic structure.

4 Results

4.1 BERT and SAE Representations

We split the dataset into a training set (80%) and a test set (20%). The training set is used to fine-tune a BERT-based classifier, while the test set is used for evaluation. The model achieves a test accuracy of 0.89 (see Table 2 for full evaluation metrics).

Metric	Value
Accuracy	0.898
Precision (Positive)	0.882
Recall (Positive)	0.92
F1-score (Positive)	0.90
Specificity (Negative Recall)	0.88
Error Rate	0.101

Table 2: Evaluation metrics for IMDB sentiment classification model

To analyze internal representations beyond BERT, we extract activations from a pretrained GPT-2 model. Specifically, we use the residual stream from layer 8 of GPT-2 small.

Let $h \in \mathbb{R}^{T \times 768}$ denote the hidden states of a given input sequence, where 768 is the model hidden dimension and $T = 128$ is the sequence length. We compute:

- GPT-2 model: GPT-2 small (12-layer transformer)
- Extracted layer: hidden state at layer 8
- Residual stream dimension: 768

These residual representations are then passed through a pretrained Sparse Autoencoder (SAE):

$$z = \text{SAE}(h)$$

where the SAE maps each 768-dimensional residual vector into a high-dimensional sparse feature space:

$$\mathbb{R}^{768} \rightarrow \mathbb{R}^{N_{\text{features}}}$$

For the GPT-2 SAE checkpoint used in this work, the feature dimension is:

$$N_{\text{features}} = [5000 \times 24576]$$

We aggregate token-level activations by mean pooling:

$$z_{\text{doc}} = \frac{1}{T} \sum_{t=1}^T z_t$$

This produces a document-level sparse representation where most feature activations are exactly zero or close to zero, reflecting the overcomplete and sparsity-inducing nature of the SAE representation (see Figure 2).

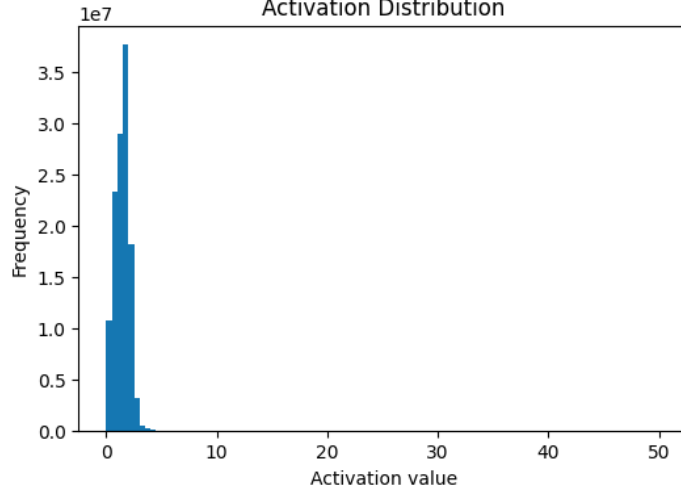


Figure 2: SAE Feature Activation Distribution

Key Observation Unlike BERT embeddings, which remain dense and low-dimensional, SAE representations decompose the GPT-2 residual stream into a large set of interpretable sparse features, enabling fine-grained analysis of semantic and stylistic structures in text.

4.2 SAE–BERT Feature Alignment Analysis

To quantify the relationship between sparse autoencoder (SAE) features and downstream classification behavior, we analyze whether SAE latent activations correlate with BERT-based predictions. Specifically, we evaluate whether individual SAE features exhibit statistically significant differences across predicted classes.

Let $Z \in \mathbb{R}^{N \times F}$ denote the document-level SAE activations, where N is the number of documents and F is the number of latent features. Let Y denote the class predictions produced by the fine-tuned BERT classifier.

Feature Discriminativeness We first compute a variance-based separability score by measuring how much each SAE feature varies across class-wise means:

$$\text{Var}(f) = \text{Var}(\mathbb{E}[Z_f|Y])$$

This captures whether a feature systematically differs across predicted classes.

Statistical Significance To assess whether feature differences are statistically meaningful, we apply Welch’s t-test for each SAE dimension:

$$p_f = \text{WelchTTest}(Z_f|Y)$$

Lower p-values indicate stronger evidence that a feature separates classes.

Predictive Contribution We further train a logistic regression classifier on SAE features to estimate predictive importance. The absolute coefficient magnitude is used as a measure of feature relevance:

$$\beta_f = |\text{LogisticRegression}(Z \rightarrow Y)_f|$$

Composite Ranking To combine complementary signals, we define a unified importance score:

$$\text{Score}_f = \text{Var}(f) + \beta_f - \log(p_f)$$

This ranking integrates class separability, predictive strength, and statistical significance.

Results We find that a subset of SAE features exhibit strong alignment with BERT predictions. Specifically:

- Top-k features show consistent agreement across variance, t-test, and logistic regression rankings.
- A subset of features achieves statistically significant class separation ($p < 0.05$).
- Logistic regression coefficients concentrate on a small number of SAE dimensions, indicating sparse predictive structure.

Quantitative summary

- Number of significant features ($p < 0.05$): **14,148**
- Top-k overlap between ranking methods ($k = 100$): **Variance–Logistic = 16, Variance–Score = 16, Logistic–Score = 50**
- Correlation between variance and logistic weights (Spearman ρ): **0.218** ($p < 0.001$)

Key Observation Using a larger evaluation set ($N = 5000$), we observe that SAE feature importance exhibits partial consistency across multiple ranking criteria. More than half of all sparse features (14,148 out of 24,576) show statistically significant class separation ($p < 0.05$), indicating widespread alignment between SAE activations and downstream classifier behavior.

Although agreement between ranking methods is not complete, moderate overlap emerges between logistic-regression-based and combined feature rankings, while variance-based rankings remain more distinct. Additionally, variance-based importance and logistic regression weights exhibit a weak but statistically significant positive correlation ($\rho = 0.218$, $p \ll 0.001$).

These findings suggest that different statistical metrics capture complementary perspectives on feature relevance. Rather than identifying a single universal notion of importance, SAE feature analysis reveals a distributed and multi-dimensional representational structure.

Interpretation Shift Initial experiments on smaller subsets suggested stronger agreement between ranking methods. However, increasing the evaluation size to $N = 5000$ revealed that feature alignment is substantially weaker and more distributed than initially expected. This indicates that SAE feature importance is sensitive to both sample size and the choice of statistical criterion.

4.3 Strong SAE Features: Semantic and Evaluative Structure

Strong and weak features are selected based on the composite importance score ranking (top-10 and bottom-10 features respectively). Based on the score the top 10 most strongly activated SAE features: 18435, 20528, 18337, 6267, 18777, 15158, 19829, 13934, 8631, and 3071. Table 3 summarizes their labels, semantic categories, and qualitative evidence derived from highly activating documents.

A clear pattern emerges across these features: SAE latents are not capturing isolated lexical or syntactic properties, but instead encode structured discourse types commonly found in film review communities. These include evaluative reasoning, aesthetic interpretation, comparative critique, and affective narrative reflection.

Feature 18435 strongly activates on long-form evaluative discourse in which users express subjective judgments about films or media content. This feature captures both positive and negative sentiment, combined with comparative reasoning and reflective interpretation. Rather than representing simple polarity, it reflects structured opinion formation grounded in narrative and character evaluation. For example:

“As kids movie it is great. For the family it just sucks. I was truly hoping for something like the Goonies which is a great film for all ages...”

Feature ID	Label	Category	Evidence
18435	Long-form evaluative media review	Topic-semantic + Sentiment	Movie reviews with emotional judgment, comparison, reflective critique
20528	Film aesthetics and cinematic analysis	Topic-semantic	Cinematic style, technicolor, voyeurism, directorial interpretation
18337	Critical comparative media analysis	Topic-semantic + Evaluative	Film comparisons, narrative critique, ideological framing
6267	Horror/cult memory-based evaluative critique	Topic-semantic + Affective memory + Evaluative	Childhood memory, horror imagery, genre comparison, writing quality critique
18777	Light evaluative entertainment discourse	Topic-semantic + Sentiment	Positive film enjoyment, comedy evaluation, comparative film judgment
15158	Narrative-driven film appreciation with structural critique	Topic-semantic + Evaluative	Plot progression, cinematography appreciation, formulaic storytelling analysis
19829	Sensory/emotional evaluative media discourse	Topic-semantic + Affective + Evaluative	Strong emotional reactions, moral/sexual framing, film judgment
13934	Low-budget film criticism and meta-commentary	Evaluative + Meta-discourse	Production critique, technical failure, sarcastic evaluation
8631	Negative evaluative entertainment rejection	Sentiment + Evaluative	Strong rejection, humor critique, dismissive audience judgment
3071	Mixed evaluative narrative review	Topic-semantic + Evaluative	Balanced critique, plot confusion, partial appreciation

Table 3: Codebook of top SAE features and qualitative interpretation

This suggests that the feature encodes a higher-level semantic concept associated with media evaluation discourse, rather than purely sentiment or lexical cues.

Feature 20528 encodes discourse focused on cinematic aesthetics and film theory, including references to visual style, directorial influence, and interpretive commentary on formal techniques. Unlike general sentiment features, this latent captures domain-specific analytical language used in film criticism. For instance:

“Brilliant use of overstated technicolor illustrates the optimistic extremes of present day Christmas ceremonies... the voyeuristic element... is unique...”

This indicates sensitivity to stylistic and formal properties of film rather than narrative content alone.

Feature 18337 captures structured comparative critique in which users evaluate films based on narrative coherence, ideological framing, and relative quality across different works. This feature reflects higher-order evaluative reasoning that integrates comparison and argumentation rather than isolated opinion. For example, users critique narrative plausibility and ideological intent, often contrasting films such as *Die Hard* or other genre benchmarks.

Feature 6267 represents a distinct discourse mode characterized by affective memory and genre-anchored interpretation. Users frequently situate their evaluation within childhood viewing experiences, particularly in horror and cult cinema contexts. This feature combines autobiographical memory with evaluative judgment and genre comparison. Unlike purely sentiment-based latents, it reflects long-term cognitive associations between early media exposure and present-day critical interpretation. It also frequently encodes normative claims about artistic quality, particularly emphasizing writing as a core determinant of cinematic value.

Feature 18777 activates on light evaluative entertainment discourse, typically associated with comedy and mainstream film appreciation. The language is characterized by accessible judgment, audience-oriented recommendations, and relatively low analytical depth compared to other features. However, it still reflects comparative evaluation across films and performers, indicating structured but less analytically dense critique.

Feature 15158 captures narrative-driven film appreciation combined with structural critique. Activations include discussion of plot progression, cinematography, pacing, and production design. While evaluative in nature, this feature emphasizes storytelling mechanics and cinematic execution, often acknowledging formulaic genre structures while still expressing overall appreciation.

Feature 19829 reflects strongly affective and morally charged evaluative discourse. It is characterized by heightened emotional language and explicit engagement with sensitive or provocative thematic content. This suggests that the feature encodes emotionally intensified interpretation of media, often tied to moral or social framing.

Feature 13934 captures meta-level criticism of low-budget or poorly executed films. The discourse often includes technical evaluation of production quality, special effects, and writing, frequently accompanied by sarcasm or dismissive commentary. This indicates sensitivity to production constraints and perceived artistic failure.

Feature 8631 represents strongly negative evaluative discourse characterized by outright rejection of media content. The language is typically dismissive, humorous, or sarcastic, often focusing on lack of entertainment value or perceived incompetence in filmmaking.

Feature 3071 encodes mixed evaluative narratives in which users simultaneously express confusion, partial appreciation, and critique. These reviews often reflect ambivalent reception, combining narrative critique with acknowledgment of some redeeming qualities.

Summary of Feature Space Structure Across all top activating features, we observe a consistent organization of SAE latents into interpretable discourse regimes. Rather than encoding surface-level lexical patterns, the SAE appears to capture higher-order semantic structures in film-related discourse. These include (i) evaluative sentiment reasoning, (ii) aesthetic and formal analysis, (iii) comparative narrative critique, and (iv) affective and memory-driven interpretation.

This suggests that sparse autoencoder features operating on transformer residual streams encode meaningful and semantically coherent units of discourse, particularly in structured domains such as film criticism.

4.4 Weak SAE Features: Semantic and Evaluative Structure

We define weak features as the lowest-ranked features according to the composite importance score. Our identified weak SAE features: 17730, 6787, 12205, 23053, 13390, 6983, 10112, 23975, 14634, 18649. Table 4 summarizes their labels, semantic categories, and qualitative evidence derived from highly activating documents.

Feature 17730 captures highly hyperbolic negative film appraisal, characterized by extreme global judgments of cinematic failure. The discourse collapses multiple dimensions of evaluation (story, editing, directing) into a single overwhelming negative verdict, often expressed through exaggerated rating scales and dismissive language. The feature reflects structured total rejection discourse rather than nuanced critique. For example, *"I'd give it a -4... Everything from the story to directing is awful... don't watch it."*

Feature 6787 reflects a templated form of extreme negative evaluation, structurally similar to Feature 17730 but with more repetitive rhetorical framing. The discourse follows a stable pattern of condemnation, often referencing prior knowledge gaps (e.g., source material awareness) to justify negativity. This suggests a learned or formulaic style of strongly negative review writing.

Feature 12205 captures low-engagement nostalgic media description, where users summarize content (often animated or older media) with minimal analytical depth. The discourse includes cultural sensitivity awareness and simple evaluative scoring, but lacks structured critique. The feature reflects descriptive recall rather than analytical judgment.

Feature 23053 is a redundant instance of hyperbolic negative appraisal, closely mirroring Feature 17730 in structure and rhetorical intensity. It reinforces the presence of a stable negative review template in the dataset, suggesting repeated encoding of extreme dissatisfaction across multiple embeddings.

Feature 13390 encodes comparative narrative and suspense analysis in classic cinema contexts. The discourse involves cross-film comparison (e.g., detective figures, war cinema), alongside interpretive

Feature ID	Label	Category	Evidence
17730	Hyperbolic negative film appraisal	Topic-semantic + Sentiment	extreme negative judgments, global film condemnation, “waste of money” framing
6787	Hyperbolic negative film appraisal (template repetition)	Topic-semantic + Sentiment	repeated extreme criticism, identical structural negativity, video game adaptation critique
12205	Nostalgic media description with low engagement	Topic-semantic + Sentiment	cartoon chase narrative summary, cultural sensitivity reference, low evaluative depth (4/10)
23053	Hyperbolic negative film appraisal (redundant instance)	Topic-semantic + Sentiment	duplicate extreme negative review, global rejection of film quality, editing/story criticism
13390	Comparative narrative and suspense evaluation	Topic-semantic + Evaluative	detective comparison, suspense scene analysis, war film interpretive framing
6983	Campy horror enjoyment and ironic appreciation	Sentiment + Entertainment	zombie gore descriptions, “bad but fun” framing, party viewing experience
10112	Biographical performance appreciation	Topic-semantic + Positive sentiment	actor performance praise, emotional biopic interpretation, character empathy and awards mention
23975	Enthusiastic fandom and serial TV appreciation	Sentiment + Fandom discourse	strong endorsement, DVD rewatch behavior, cancellation regret and fan loyalty
14634	Mixed evaluative action film critique	Topic-semantic + Evaluative	technical assessment of plot and acting, balanced praise and criticism, action genre framing
18649	Emotion-driven negative character judgment	Sentiment + Affective reaction	strong dislike of characters, disgust-based reaction, memory-linked emotional recall

Table 4: Codebook of weak SAE features and qualitative interpretation

evaluation of narrative tension and thematic depth. Unlike purely emotional reviews, this feature reflects moderate analytical reasoning about storytelling and genre conventions.

Feature 6983 captures campy horror appreciation and ironic enjoyment of low-budget media. Despite recognizing poor production quality, users express enjoyment through humor, group viewing (“party tape” contexts), and exaggerated gore descriptions. This reflects a dual-layer sentiment structure: criticism coexisting with entertainment value.

Feature 10112 represents biographical film appreciation grounded in performance evaluation and emotional engagement. The discourse highlights actor transformation, narrative inspiration, and character empathy, often praising acting intensity and awards recognition. The feature encodes strong positive sentiment combined with interpretive admiration.

Feature 23975 captures fan-driven serial television enthusiasm, characterized by emotional attachment, rewatch behavior, and frustration over cancellation. The discourse reflects community-based valuation, where narrative quality is reinforced by fandom identity and continuity expectations.

Feature 14634 reflects mixed evaluative discourse in action cinema, combining technical critique (plot, writing, acting) with selective praise (setting, performance). The feature encodes balanced judgment rather than polarity extremes, suggesting structured but moderate analytic engagement.

Feature 18649 encodes emotion-driven negative character evaluation, where users express strong affective reactions such as disgust, dislike, or discomfort. The discourse is less about cinematic structure and more about moral or emotional rejection of characters and experiences, often reinforced by memory recall.

Summary of Weak Feature Space Structure Across the weak-feature set, we observe a consistent shift away from structured analytical reasoning toward extreme affective expression, templated

review patterns, and emotionally driven judgment. Unlike the strong features, which encode domain-specific critique and comparative reasoning, weak features are characterized by: (i) Hyperbolic global evaluations (e.g., total film failure narratives) (ii) Repetitive or templated review structures (iii) Nostalgic or low-engagement descriptive summaries (iv) Strong emotional reactions (disgust, enjoyment, frustration) (v) Reduced analytical depth and limited conceptual abstraction.

Interestingly, some weak features still contain evaluative language; however, this evaluation is typically non-structured, affect-dominant, or formulaic, indicating that the model captures sentiment intensity more than interpretive reasoning in these dimensions.

Overall, the distinction between strong and weak features suggests that SAE importance rankings align with differences in discourse structure, where higher-ranked features tend to reflect more structured evaluative reasoning compared to lower-ranked features which more often exhibit templated or affect-driven expression.

4.5 Ambiguous and Mixed Features

Some SAE features exhibit mixed characteristics, combining both semantic and structural signals. These features do not clearly align with either high-level evaluative discourse or low-level affective templates. This suggests that SAE latents may represent partially disentangled factors of discourse rather than strictly separable categories.

5 Discussion

5.1 Multi-View Structure of SAE Feature Importance

Our results suggest that sparse autoencoder features extracted from GPT-2 residual streams exhibit a multi-view structure with respect to downstream classification behavior. While a large proportion of features (14,148 out of 24,576) show statistically significant class separation, the agreement between different importance measures is limited.

In particular, variance-based ranking, logistic regression coefficients, and combined statistical scores identify partially overlapping but distinct sets of top features, with limited overlap between variance- and logistic-based rankings. This indicates that different analytical criteria capture distinct aspects of the SAE representation space rather than a unified notion of “importance.”

5.2 Weak but Significant Alignment with Classifier Behavior

Despite this divergence, we observe a statistically significant but weak-to-moderate positive correlation between variance-based feature importance and logistic regression weights ($\rho = 0.218$, $p \ll 0.001$). This suggests that while there is a global tendency for features that separate classes to also contribute to linear prediction, this relationship is not strong at the level of individual features.

This finding is consistent with the hypothesis that SAE representations are highly overcomplete and distributed, such that task-relevant information is spread across many partially redundant features.

5.3 Evaluation of the Feature-Semantics Hypothesis

A central hypothesis of this work was that sparse latent features most strongly aligned with classifier outputs would encode higher-level semantic concepts, whereas weakly aligned features would capture lower-level linguistic or stylistic structure.

Our qualitative analysis provides partial support for this hypothesis. Highly ranked features consistently activated on semantically rich evaluative discourse, including comparative film critique, aesthetic analysis, narrative interpretation, and structured opinion formation. These features reflect abstract discourse-level reasoning rather than isolated lexical triggers.

In contrast, weakly ranked features more frequently captured repetitive rhetorical templates, emotionally exaggerated reactions, nostalgic descriptive summaries, and formulaic evaluative expressions. While these features were not purely grammatical or syntactic in nature, they generally exhibited reduced conceptual abstraction and weaker semantic organization.

This suggests that classifier-aligned sparse features tend to correspond to higher-order semantic organization, while weakly aligned features more often reflect shallow stylistic or affective regularities. However, the distinction is not absolute, indicating that semantic interpretability exists along a continuum rather than as a binary separation.

5.4 Implications for Interpretability

These results challenge the assumption that sparse autoencoder features correspond to a single, stable ranking of semantic importance. Instead, SAE latents appear to encode a rich but non-uniform feature space where interpretability depends strongly on the chosen analytical lens.

In this setting:

- Variance captures discriminative structure across classes,
- Logistic regression captures predictive linear directions,
- t-tests capture distributional shifts,
- Combined scoring integrates these but does not reconcile them.

Therefore, interpretability of SAE features should be understood as multi-dimensional rather than scalar.

5.5 Relation to BERT Classifier Signals

Importantly, the observed partial alignment between SAE features and BERT predictions suggests that GPT-2 residual stream structure contains signals relevant to downstream classification tasks. However, this alignment is indirect and distributed, rather than being concentrated in a small subset of highly predictive features.

This supports the view that SAE decomposition reveals latent semantic structure in transformer representations, but does not yield a direct one-to-one mapping with classifier decision boundaries.

5.6 Limitations and Future Directions

While the results demonstrate meaningful alignment between sparse autoencoder features and downstream classifier behavior, several limitations should be noted.

First, the current analysis primarily relies on linear and univariate statistical measures, including variance-based separation, Welch’s t-test, and logistic regression coefficients. These approaches may not fully capture nonlinear or higher-order interactions between SAE features and classifier predictions. As a result, some complex representational dependencies in the SAE feature space may remain unexplored.

Second, this study uses GPT-2 small as the underlying generative model for residual stream extraction. Although GPT-2 provides a well-studied and computationally tractable setting, it is significantly less capable than modern large language models such as Gemma or other more recent transformer architectures. Consequently, the representational richness of the extracted residual stream may be limited compared to more advanced models.

Third, the analysis is constrained by a relatively small set of qualitative inspections (top-k activating documents per feature). While this provides interpretability insights, it does not scale to fully characterize the entire SAE feature space.

Future work could address these limitations in several ways. First, nonlinear analysis methods, such as mutual information estimators or kernel-based methods, could be used to better capture complex dependencies between SAE features and classifier behavior. Second, extending the analysis to larger and more capable language models would allow evaluation of whether the observed structural patterns generalize across model scales. Third, qualitative analysis could be partially automated or augmented using clustering methods or embedding-based grouping of high-activation documents, enabling more systematic exploration of feature semantics at scale.

6 Conclusion

This work investigates whether sparse autoencoder (SAE) features extracted from GPT-2 residual streams align with downstream classifier behavior. We propose a cross-model analysis framework that relates SAE latent activations to BERT-based predictions using multiple complementary statistical criteria, including variance-based separation, Welch’s t-test, and logistic regression coefficients.

Our results show that SAE feature importance is not consistent across ranking methods, revealing a multi-view structure in the learned representation space. While a large fraction of features exhibit statistically significant class separation, agreement between ranking criteria remains limited, and feature-level correlations are weak. This indicates that no single importance metric fully captures the structure of the SAE feature space.

Despite this inconsistency, qualitative analysis reveals a stable semantic pattern: strongly aligned features tend to capture structured evaluative and discourse-level reasoning, whereas weakly aligned features more often reflect templated, affect-driven, or less compositional language patterns. This suggests that interpretability emerges not from individual ranking signals alone, but from the interaction between statistical selection criteria and qualitative inspection.

Overall, our findings support the view that sparse autoencoder representations encode distributed and multi-faceted semantic structure that partially reflects classifier decision boundaries, but does not reduce to a single predictive decomposition. This highlights both the promise and the limitations of SAE-based interpretability for understanding transformer representations.

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